Case Vignettes:
Urologic Emergencies

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Learning Objectives

1. Define acute urinary retention and describe its presentation, treatment, and possible sequelae.
2. Define obstructive pyelonephritis and describe its presentation, diagnosis, treatment, and possible sequelae.
3. Define priapism and describe its presentation, diagnosis, treatment, and possible sequelae.
4. Define penile fracture and describe its presentation, diagnosis, treatment, and possible sequelae.
5. Define paraphimosis and describe its presentation, diagnosis, treatment, and possible sequelae.
6. Define testicular torsion and describe its presentation, diagnosis, treatment, and possible sequelae.
7. Define Fournier's gangrene and describe its presentation, diagnosis, treatment, and possible sequelae.

Case 1 – History

A 35-year-old patient presents with a persistent erection lasting over 5 hours.

He reports taking a new medicine which he thinks may have contributed to his condition. After further pressing, he admits that the ‘medication’ is actually cocaine.

The erection has been painful starting 15 minutes after he noticed it.

PM/SHx

- HTN
- Polysubstance abuse
- No surgeries

Questions

1. *What is the most likely diagnosis?*

2. *What are the 2 different types of priapism? How does each occur?*

3. *How do you differentiate between these 2 types?*

Questions

1. *What are the potential sequelae of both types of priapism?*

2. *How is each type of priapism treated?*

Case 2 - History

A 78-year-old patient with severe dementia presents with significant agitation.

He has been like this for the last 12 hours. A caregiver at his nursing home noted that he wasn’t eating and became concerned.

PM/SHx

- HTN
- DM
- Severe dementia
- Hypothyroidism
- HL
- Sarcoidosis
- h/o CLL
- BPH

- h/o OLT
- Bowel resection
- TURP
- Toe amputation

Meds/SHx/All

- NKDA

- Polypharmacy (25 pills daily)

- Non-communicative
- Non-ambulatory
- Lives in nursing home

Exam

- Agitated
- Temp 98.9F
- BP 186/87
- HR 111
- Sat 99% RA

- HEENT
- Pulm – clear, bibasilar rales
- CV – sinus tach
- Abd – soft, NT, ND
- GU – will not let you examine

- After much effort, you are able to lift his gown with his caregiver …



Questions

1. *What is the diagnosis?*

2. *What is the etiology?*

3. *What other question could you have asked during the history that may have suggested this diagnosis?*

4. *How do you treat this condition?*

Case 3 - History

A 10-year-old child is brought into the emergency room with a 2-hour history of an exquisitely tender left hemi-scrotum.

He was playing tetherball and was hit in the groin when he was not paying attention.

He has vomited once since that incident. He has been able to void spontaneously. Denies f/c.

PM/SHx

- None

- Hypospadias repair as an infant

Meds/SHx/All

- NKDA
- Adderall XR
- 4th grade
- Normal development

Exam

- Agitated
- Temp 98.9F
- BP 186/87
- HR 111
- Sat 99% RA

- HEENT
- Pulm – clear
- CV – sinus tach
- Abd – soft, NT, ND
- GU – will not let you examine

- After much effort, you are able to lift his gown with his mother …



Questions

1. *What do you see on the exam that tips you off to the diagnosis?*

2. *If the exam is not clear, what else can you order?*

Questions

1. *What is the anatomic abnormality associated with a testicular torsion?*

1. *What is the treatment for this patient?*

2. *What is the critical time period during which this must be corrected?*





Case 4 - History

A 27-year-old patient comes into the emergency room with a 4-hour history of left flank pain and fever

The pain awoke her at night and is severe. It comes and goes but when it is present, she cannot find a comfortable position.

She has vomited once since it started. She denies GH.

PM/SHx

- None

- C-section

Meds/FHx/SHx/All

- NKDA
- OCM
- Works as a teacher
- Married with one child – healthy
- FHx: significant for nephrolithiasis

Exam

- Agitated
- Temp 103.1F
- BP 98/64
- HR 111
- Sat 99% RA

- HEENT
- Pulm – clear
- CV – sinus tach
- Abd – (+) L CVA, benign
- Ext – cool, difficult to palpated pulses

Questions

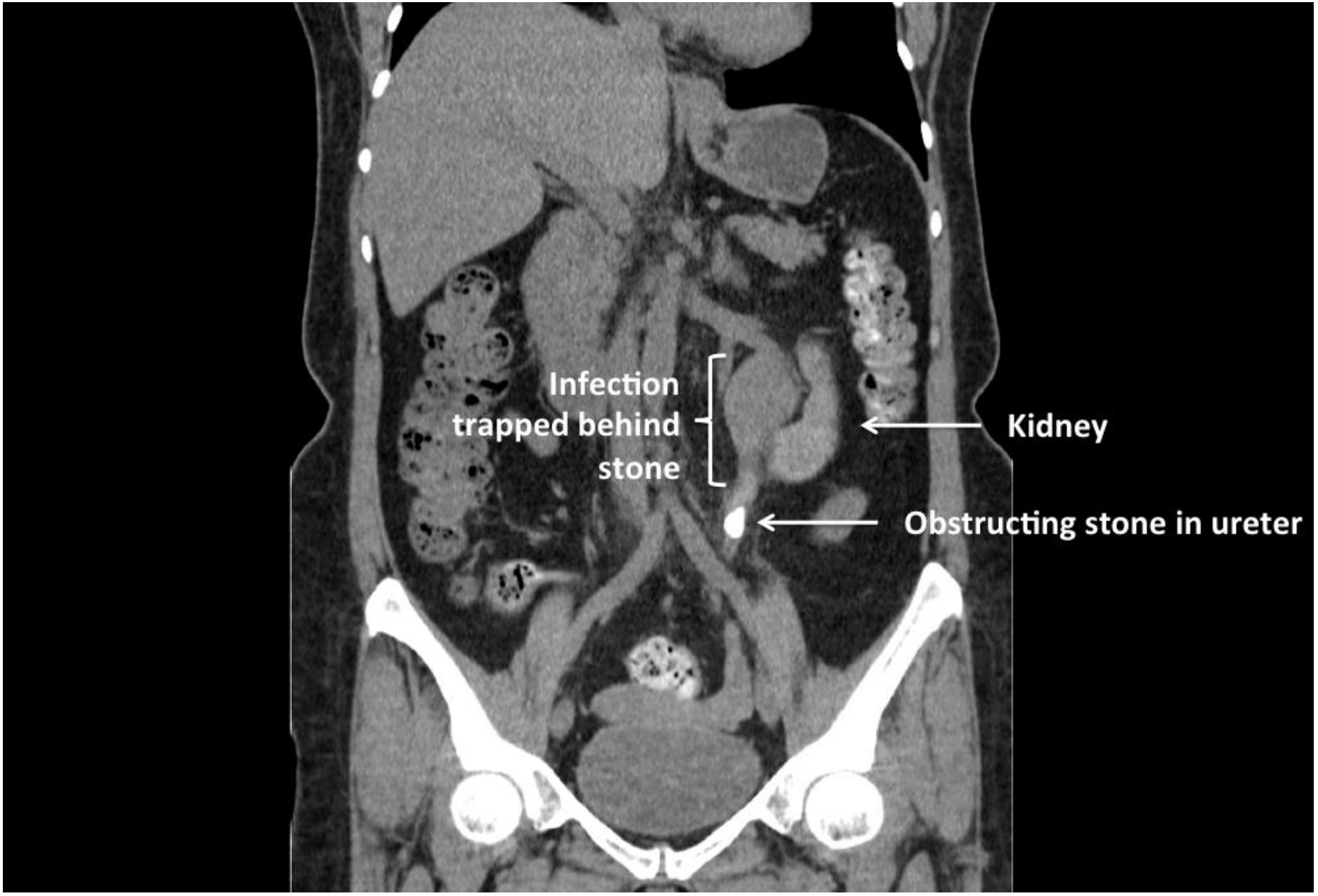
1. *What do you order at this point?*

- *UA – 250+ RBC, (+) LE, (+) nit*
- *Urine culture pending*
- *Blood culture pending*
- *CBC: Hg 12.1, WBC 25.2 (left shift)*
- *BMP: BUN/Creat 26/1.5*

Questions

1. *What is your working diagnosis?*

2. *What test do you order at this point?*


Infection
trapped behind
stone
Kidney
Obstructing stone in ureter

Questions

1. *How do you manage this patient?*

2. *What is the danger of not managing this patient expeditiously?*

Case 5 - History

A 63-year-old patient comes into the emergency room with severe pain and swelling in the perineal region for 2 days with associated fever and malaise.

Sudden onset of intense pain in the perineal area two days ago, which progressively worsened. He noticed swelling and redness spreading around the scrotum and perineum.

He also complains of fever, chills, and feeling generally unwell. No history of trauma reported.

PM/SHx

- Type 2 Diabetes Mellitus (diagnosed 10 years ago, poorly controlled)
- Hypertension
- Obesity (BMI 32)

Meds/FHx/SHx/All

- NKDA

- Metformin, Lisinopril

- Smokes 1 pack/day for 20 years
- Occasional alcohol use
- Lives with family, no recent travel or sick contacts

Exam

Vital Signs:

- Temp: 38.7° C (101.7° F)
- HR: 110 bpm
- BP: 100/65 mmHg
- RR: 30/min
- SpO2: 98% on room air

General: Appears toxic and distressed due to pain

- Perineal region: Marked swelling and erythema of the scrotum and perineum
- Skin is warm, tender, and tense with areas of fluctuance
- Crepitus felt on palpation
- Foul-smelling, necrotic tissue visible with some purulent discharge

Genitalia:

- Testes difficult to palpate due to swelling
- Abdomen: Soft, non-tender, non-distended
- Other systems: No remarkable findings

Questions

1. *What do you order at this point?*

- *CBC: Leukocytosis (WBC 18,000/mm^3) with left shift*
- *Blood glucose: 320 mg/dL (hyperglycemia)*
- *CRP elevated*
- *Serum creatinine mildly elevated (1.6 mg/dL)*
- *Blood cultures: Pending*
- *Urinalysis: No infection noted*
- *Electrolytes: metabolic acidosis*

Questions

1. *What is your working diagnosis?*

2. *What test do you order at this point?*


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Questions

1. *How do you manage this patient?*

2. *What is the danger of not managing this patient expeditiously?*